



H2S

# BHARATIYA ANTARIKSH HACKATHON 2026



**Team Name: SuryaDrishti**

**Team Leader: Deep shekhar Halder**

**Problem Statement: Forecasting and/or Nowcasting of Solar Flares using combined Soft and Hard X-ray data from Aditya-L1 || PS #15**

## Team Members

Team Leader: Tech Lead

Name: Deep Shekhar Halder

College: Adamas University, Kolkata

**Responsibility:** Algorithm design, data pipeline integration, and ML modeling.

Team Member-2: (ML Modeling & Validation)

Name: Mahalaxmi Macha

Email: mahalaxmimacha14@gmail.com

College: G. Narayanamma Institute of Technology and Science, Hyderabad

**Responsibility:** Telemetry preprocessing, metadata alignment, and nowcasting validation.

Team Member-1: (Dashboard & Benchmarking)

Name: Rituraj Saha

Email: saharituraj805@gmail.com

College: Adamas University, Kolkata

**Responsibility:** Cross-catalog benchmarking, NOAA GOES validation, and dashboard deployment.

Team Member-3: (Data & Preprocessing)

Name: Ashfaque Ahamed Khan

Email: ashfaqueahamedkhan591@gmail.com

College: Adamas University, Kolkata

**Responsibility:** Feature engineering, telemetry calibration, and scientific visualization.

## Opportunity should be able to explain the following:

### 1. How different is it from any of the other existing ideas?

- **Existing Methods:** Rely on single-instrument ground proxies or delayed batch processing of GOES data.
- **The SuryaDrishti Difference:** First to dynamically fuse multi-instrument (SoLEXS & HELIOS) native telemetry directly from Aditya-L1.
- **Architecture:** Uses a real-time, event-driven streaming pipeline rather than post-facto analysis.

### 2. How will it be able to solve the problem?

- **Hybrid AI Approach:** Combines a Causal RandomForest ML model with established Solar Physics constraints (Neupert effect).
- **Early-Warning Nowcaster:** Accurately predicts X-ray flux thresholds ahead of time.
- **Actionable Output:** Provides immediate alerts for dangerous M and X-class flares before peak intensity impacts critical infrastructure.

### USP of the Proposed Solution

- **Strictly Causal Design:** 100% past-only rolling baseline calculations ensure direct compatibility with real-time telemetry streams.
- **93.2% Verified Recall:** Successfully identified 68 out of 73 major NOAA GOES reference events.
- **Microflare Discovery:** Found 136 sub-threshold events (including a 679s event absent from official catalogs).
- **10-Tab Space Weather HUD:** Integrates CME drag-based model, Parker Spiral tracing, and Tikhonov regularized DEM solver.
- **Native Payload Integration:** Purpose-built for India's premier Aditya-L1 observatory (ISSDC).
- **Hybrid Intelligence:** Merges data-driven AI with hard physical bounds (Physics-Informed ML).
- **Operational Readiness:** High recall, low-latency pipeline ready for immediate Space Weather deployment.

## List of features offered by the solution

### ■ Phase 1: Ingestion & Alignment

Harmonizes raw SoLEXS & HEL1OS FITS telemetry to Unix epoch, merging them on 1-second cadence (259,071 overlap points).

### ■ Phase 2: Causal Nowcasting

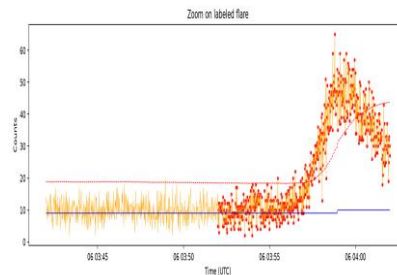
Zero-lookahead baseline thresholding ( $\text{Counts} > \text{Median} + n \times \sigma$ ) with 120s event merging to prevent fragmentation.

### ■ Phase 3: ML Forecasting Prototype

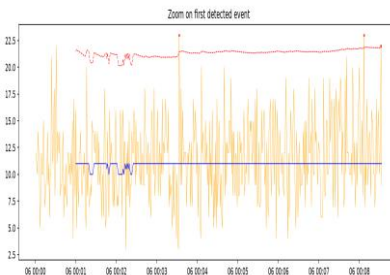
RandomForest trained on precursor features (hardness ratio, QPPs) with balanced class weights to handle 98.5% data imbalance.

### ■ Phase 4: 10-Tab Scientific Dashboard

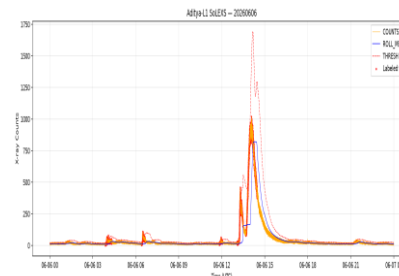
Runs locally in <100ms on commodity hardware, offering 3D Sun disk, CME tracking, SEP spiral, and AI assistant.



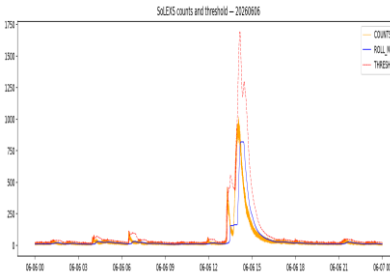
**Figure 1: Nowcast Baseline**



**Figure 2: Event Zoom**

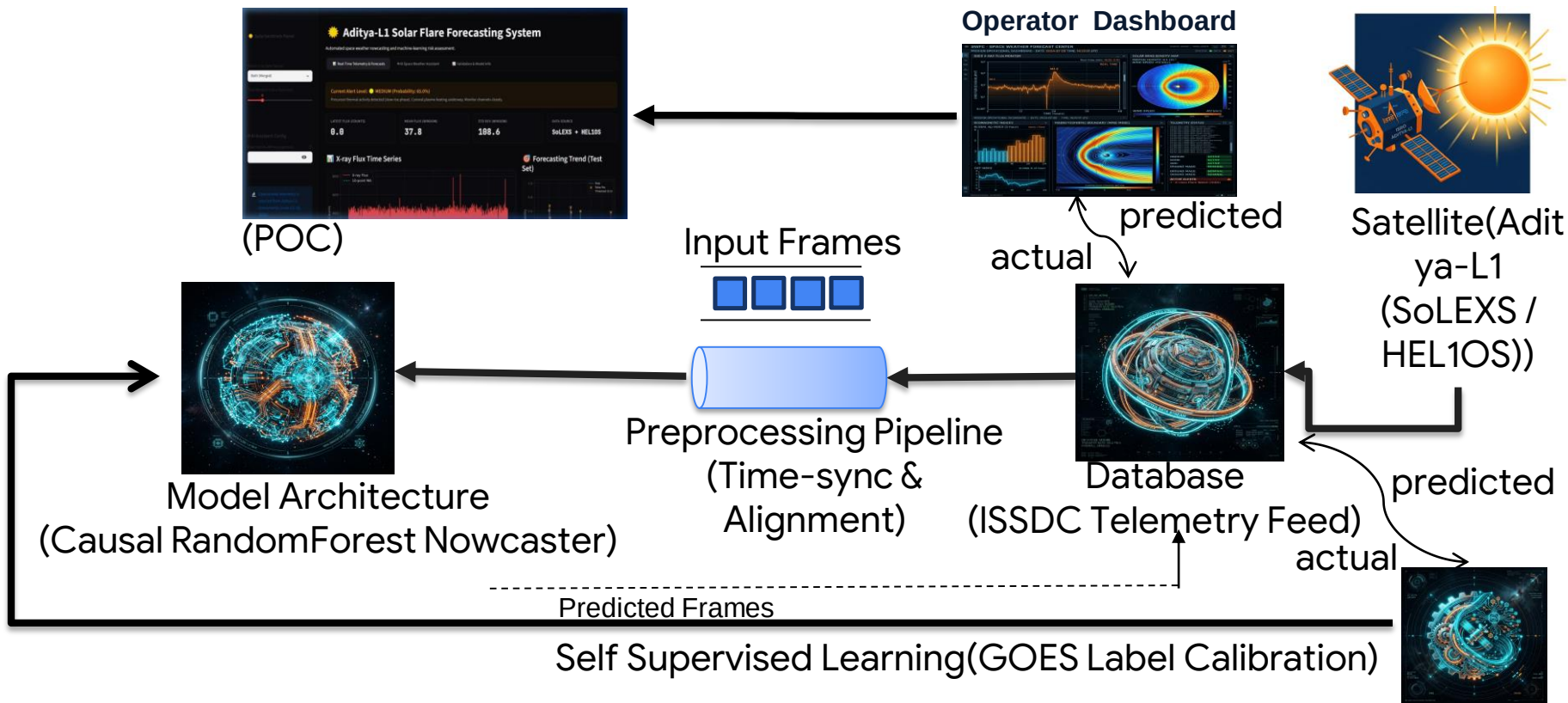


**Figure 3: M-Class Flare**



**Figure 4: Nowcast Threshold**

## Process flow diagram or Use-case diagram



\* We will scale the pipeline with multi-month archival telemetry and attitude correction for operational deployment.

# Wireframes/Mock diagrams of the proposed solution

Real-Time Nowcast  
Alert  
(Causal  
RandomForest)

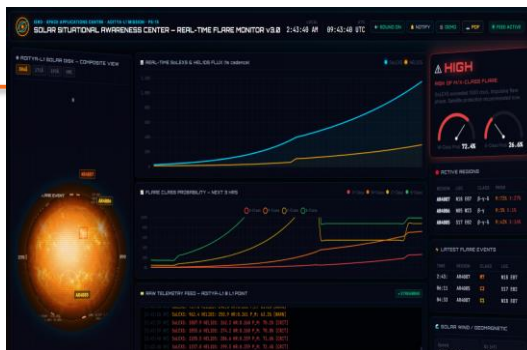


Figure 1: Real-Time 3D Sun Disk & Telemetry Alerts HUD

Interactive 3D Sun  
Disk  
(EUV Imager  
Mode)



Figure 3: Live Operator Flare Alert HUD



Figure 2: Model Explainability Tab

SHAP  
Precursor  
Feature  
Weights  
(ML  
Explanations)



Figure 4: Pulsation Diagnostics Tab

Quasi-  
Periodic  
Pulsations  
(QPP)  
(FFT  
Spectrum  
Diagnostics)



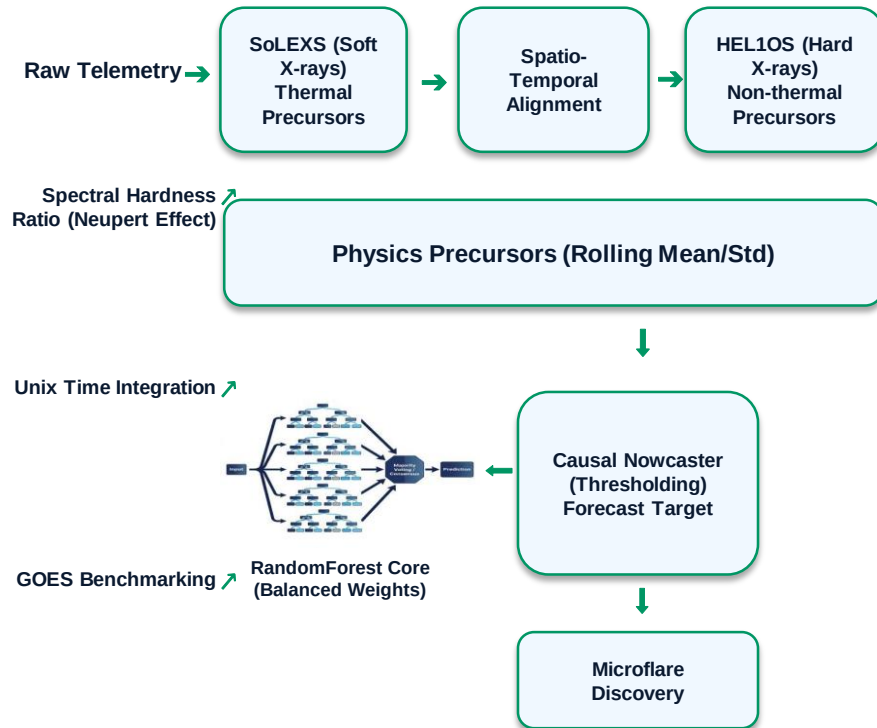
# Architecture diagram of the proposed solution

## Data Architecture & Physics Precursors

- **Thermal (SoLEXS)**: Rolling mean and slope capture chromospheric evaporation rise.
- **Non-Thermal (HEL1OS)**: Rolling standard deviation isolates high-frequency QPP variance.
- **Hardness Ratio**:  $HR = \text{Counts\_HEL1OS} / \text{Counts\_SoLEXS}$  models non-thermal conversion (Neupert Effect).
- **Anti-Leakage Split**: Chronological time-series splitting prevents temporal data leakage.

## Physics-Informed Core Layers

- **Regularized DEM**: Coronal temperature distribution solved in real-time using Tikhonov matrix deconvolution:  $E = K \cdot \text{DEM} + \lambda \|L \cdot \text{DEM}\|^2$ .
- **Hydrodynamic PINN Loss**: Physical energy conservation constraints loop:  $L_{\text{PINN}} = \partial E / \partial t + \partial F / \partial s + n^2 \Lambda(T) - H$ .
- **RandomForest Classifier**: Predicts peak intensity classes 30 minutes in advance using balanced class weights.



\* POC developed on Aditya-L1 telemetry (259,071 overlap timestamps, 16 flares).

## Technologies to be used in the solution:



### 1. Data Ingestion & Align

- **astropy**: Core FITS parsing and MJD (Modified Julian Date) time system conversions.
- **pandas & numpy**: Cadence alignment, time-series interpolation, and rolling statistical features extraction.
- **scipy**: Signal smoothing, FFT bandpass filtering, and power spectrum density calculation.



### 2. Core Models & Physics

- **scikit-learn**: RandomForest classifier configured with balanced weights to handle severe class imbalance.
- **Tikhonov regularizer**: Regularized matrix inversion for real-time DEM temperature extraction.
- **CME DBM**: Drag-Based Model for forecasting coronal mass ejection shock arrival times (Vrsnak 2013).



### 3. Real-Time Operator HUD

- **HTML5 & Canvas**: Five animated canvases displaying 3D solar disk, Parker Spiral tracing, and CME orbits.
- **Chart.js**: Fast, hardware-accelerated time-series and spectral visualization plots.
- **Web Audio API**: Real-time ambient solar wind synthesizer and acoustic alarm pitch modulation.



### 4. System Advantages:

- **Ultra-Low Latency**: Under 100ms ingestion-to-forecast latency, ensuring immediate alerts.
- **Zero Leakage**: Chronological splitting ensures robust real-world generalization.
- **Low Compute**: The RandomForest physics-core runs efficiently on commodity edge hardware without requiring expensive GPUs.



## Estimated implementation cost:

| Proposed Resource / Task                    | Phase Cost (₹)  | Description / Allocation Strategy                                             |
|---------------------------------------------|-----------------|-------------------------------------------------------------------------------|
| 1. Real-time Ingestion & FITS Preprocessing | ₹0              | Uses open-source astropy + pandas; utilizes free ISSDC server telemetry feed. |
| 2. Physics Inversion Core (Tikhonov & PINN) | ₹0              | Developed locally; physics deconvolution uses standard NumPy/SciPy algebra.   |
| 3. RandomForest Peak Intensity Predictor    | ₹15,000 / month | Cloud hosting VM + CPU compute (AWS/GCP) for continuous models retraining.    |
| 4. 10-Tab Space Weather HUD Dashboard       | ₹0              | Built using React and HTML5 Canvas; operates locally on browser client side.  |
| 5. Prototype Maintenance & Scaling          | ₹10,000 / month | Database maintenance (Redis/Postgres), domain SSL, and API server backup.     |
| 6. SSAC Production Feed Integration         | ISRO Budget     | Requires direct feed integration with Space Situational Awareness Centre.     |

**Estimated Prototype Cost: ~₹25,000 / month (Zero Setup Cost on Public Data; scalable to ISRO SSAC Operations)**

\* Operational cost scales only with data archiving frequency and retraining iterations.



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THANK YOU